

Physics Assignment 1

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1. A $12\ \Omega$ resistor is connected to a 24 V supply. Find the current through it. [1]

By Ohm's law, $I = V / R$.

$$I = V / R = 24\text{ V} / 12\ \Omega = 2\text{ A}$$

2. Two $2\ \Omega$ resistors are connected in parallel across a 5 V battery. Find I_1 . [1]

In a parallel connection both resistors see the full battery voltage (5 V). So the current through the first resistor is:

$$I_1 = V / R = 5\text{ V} / 2\ \Omega = 2.5\text{ A}$$

(Each branch carries 2.5 A, so the total current drawn from the battery is $2.5 + 2.5 = 5\text{ A}$.)

3. A $6\ \Omega$ resistor is connected to a 12 V battery. A particle of mass 2 kg experiences a force equal (in newtons) to the current in the resistor. Find its acceleration. [2]

First find the current, then use Newton's second law ($a = F / m$).

$$\begin{aligned} I &= V / R = 12\text{ V} / 6\ \Omega = 2\text{ A} & \rightarrow & F = 2\text{ N} \\ a &= F / m = 2\text{ N} / 2\text{ kg} = 1\text{ m/s}^2 \end{aligned}$$

4. What is meant by overclocking? [2]

Overclocking means running a component (usually a CPU or GPU) at a higher clock frequency than the speed the manufacturer rated it for, so it executes more cycles per second and performs faster. Think of a car engine: the maker sets a safe redline, and overclocking is pushing past that redline to get more power. The trade-off is that the chip draws more power, runs hotter, and can become unstable or wear out sooner, so it usually needs better cooling and a higher supply voltage to stay stable.

5. What is electromagnetism? Explain. [2]

Electromagnetism is the branch of physics that studies the interaction between electric charges and magnetic fields, and how the two are really one connected phenomenon.

The core ideas:

- A moving charge (an electric current) produces a magnetic field around it.
Example: a current-carrying wire deflects a nearby compass needle (Oersted's experiment).

- A changing magnetic field produces an electric current. Example: moving a magnet in and out of a coil induces a voltage (Faraday's law of electromagnetic induction).

So electricity and magnetism are two sides of the same coin, unified by Maxwell's equations. It is the principle behind motors, generators, transformers and electromagnets.

6. Which correctly describes the magnetic field near a long straight current-carrying wire? [1]

(d) The field consists of concentric circles centred on the wire. The field lines form closed loops around the wire, and their direction is given by the right-hand thumb rule (thumb along the current, fingers curl in the field direction).

7. At the instant of a short circuit, the current in the circuit will _____ [1]

(c) Increase heavily. A short circuit bypasses the load, so the resistance drops to almost zero. By $I = V / R$, as $R \rightarrow 0$ the current shoots up to a very large value (which is why fuses and breakers trip).